

Aravind Sundaresan

Systems Engineer — LLM Inference Engine Internals & GPU Performance

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Systems Engineer with 9 years of software engineering — 6 at **Microsoft** and **Amazon** building production distributed platforms, 3 on independent LLM-inference and GPU performance research. Upstream contributor to **vLLM** and **SGLang** core internals — open PRs on the v1 scheduler, disaggregated prefill/decode memory management, and Marlin-quantization fallbacks for speculative decoding and new model architectures. Hands-on GPU performance work (Nsight Systems profiling, roofline reasoning, tiled CUDA kernels) and a hardware-aware disaggregated-serving system with in-flight FP8 KV-cache quantization, benchmarked on real GPUs.

UPSTREAM OPEN-SOURCE CONTRIBUTIONS — VLLM & SGLANG

Open (not yet merged) as of Sept 2026; #51574 is co-authored

- **vLLM #51574** — Filed the original bug (#41951: preempted requests lose re-admission priority in the v1 `PriorityRequestQueue`) and wrote the first fixes (#41952, #56843; closed as stale/consolidated); co-author (unit tests) on the consolidated fix in #51574, which is open under another contributor.
- **vLLM #56964** — Fixed a crash where quantization-only kernel backends (Marlin) could not execute the unquantized draft-model layers used in speculative decoding (MTP); added a graceful fallback to auto for unquantized layers.
- **SGLang #39741** — Fixed a crash loading Qwen3.5 GDN hybrid models where Marlin's tile-divisibility requirement ($N \bmod 64 = 0$) was silently violated by an unquantized projection layer; added a tile-divisibility check with fallback to the unquantized linear method.
- **SGLang #39623** — Diagnosed and fixed unbounded memory growth in prefill/decode disaggregation from orphaned ZMQ bootstrap-room state; added admission-guard validation and an $O(1)$ min-heap TTL sweeper (replacing an $O(N)$ dictionary scan), with parity across three transfer backends.
- **SGLang #37067** — Refactored `RadixCache.total_size()` from $O(N)$ DFS traversal to $O(1)$ using an existing size invariant — a code-hygiene/API-consistency improvement (confirmed via maintainer review that this path is not in the production hot loop), with unit test coverage across all cache mutation paths.

GPU SYSTEMS & INFERENCE PROJECTS

HeteroDisagg — Hardware-Aware Disaggregated Serving

github.com/Aravind0403/HeteroDisagg

Real A100 (prefill) + RTX 3090 (decode) testbed, Vast.ai · Explicit empirical-vs-projected demarcation

- Built HeteroKVConnector: asymmetric-TP KV-cache re-sharding across mismatched prefill/decode ranks (TP=4 → TP=1) with in-flight FP8 block quantization — **99.1–99.7% output cosine similarity** and $\Delta\text{PPL} \leq 0.0018$ vs. FP16 (Llama-3-8B, 512–4096 tokens), **2×** wire-payload reduction (128MB → 64MB).
- Measured isolated GPU-kernel cost of quantization/resharding honestly: slower than uncompressed FP16 in raw compute time (+11–32ms depending on concurrency) — the real payoff is wire-bandwidth reduction under network-constrained transfer, not kernel speed; production-fabric (NVLink/RoCE) throughput stated as analytical projections, not measured.

Multi-Node-Inference-Lab — CUDA Profiling & Scheduler Internals

github.com/Aravind0403/multi-node-inference-lab

CUDA C++, Nsight Systems, vLLM/SGLang internals

- Occupancy analysis identifying the 4-warps/block throughput plateau on a memory-bandwidth-bound kernel; shared-memory-tiled matmul at $1.6\text{--}2.5\times$ speedup over naive (correctness-verified against baseline); float4 vectorization limits under bandwidth saturation.
- Comparative internals analysis of vLLM's PagedAttention/block-hash scheduler vs. SGLang's RadixAttention/radix-tree scheduler.

Aether Control — vLLM Serving Control Plane & GRPO Post-Training

github.com/Aravind0403/aether-control-llm-infra

vLLM, HuggingFace TRL, DeepSpeed ZeRO-3, Kubernetes

- PagedAttention + Chunked Prefill cut short-query P99 TTFT by 88.9%; root-caused a real 81–93% throughput-collapse incident under aggressive VRAM optimization to an upstream data-validation gap, fixed at the correct layer.
- GRPO post-training (critic-free, DeepSpeed ZeRO-3): gains verified on two independent hardware runs (GCP L4, bare-metal RTX 4090).

PROFESSIONAL EXPERIENCE

Founder & LLM Inference / ML Infrastructure Engineer · Anavind Labs

Nov 2023 — Present

Independent Research Practice · Hyderabad, India

- Founded and operate an independent AI-systems research practice focused on GPU kernels and LLM inference — the upstream open-source contributions, benchmarked serving/training systems, and CUDA kernel studies detailed above are its output; every result published with reproducible, disclosed-variance benchmarks ([arXiv:2606.07248](https://arxiv.org/abs/2606.07248)).

Software Engineer II, Platform & Infrastructure · Microsoft R&D India

Aug 2021 — Nov 2023

Hyderabad, India

- Architected a globally distributed metadata and governance platform across 17,000+ microservices at 99.99% availability; Redis-based adaptive concurrency control eliminated 50+ production retry storms.
- Built a C++ Test-in-Production telemetry framework — function entry/exit hooks, state-transition recording, anomaly surfacing — adopted across 7–10 teams (the entire Windows India chapter) as a CI/CD certification gate, reaching 90% instrumentation coverage.
- Implemented a 4-stage Azure DevOps validation pipeline (cross-compile → environment setup → E2E test execution → gate decision), raising ARM64 hardware-compatibility coverage to 80%+ across Windows 11 builds and preventing regressions from reaching GA.
- Built an asynchronous data pipeline (Azure Data Factory → Kusto/Azure Data Explorer) syncing transactional SQL updates for near-real-time compliance reporting and analytical querying across the 17,000+ microservice platform.

Software Development Engineer, Systems & Device Infrastructure · Amazon Development Centre

Sep 2017 — Aug 2021

Chennai, India

- Engineered a highly concurrent Java device-scheduling engine orchestrating 24/7 deterministic integration and stress test suites across 50+ physical device categories (Echo, Fire TV, Fire tablets), achieving 98.4% scheduler uptime — category-aware job routing, deterministic pre/post-test state management, automatic ADB-based recovery.
- Saved 1,000+ developer debugging hours annually by authoring unified diagnostic APIs/tooling (logcat, bugreport, dumpsys, dmesg aggregation) and self-healing physical-infrastructure recovery protocols, adopted organically by 7+ Amazon device teams.
- Built 40+ deterministic hardware-software integration test cases covering critical device scenarios.
- Accelerated Alexa Comms build qualification via automated CI/CD validation workflows; built Python/Bash release automation scripts coordinating artefact promotion across device categories.

TECHNICAL SKILLS

- **GPU Performance & CUDA:** CUDA C++, Nsight Systems profiling, occupancy analysis, shared-memory tiling, memory-hierarchy reasoning.
- **Quantization:** AWQ/GPTQ, fp8, Marlin kernel constraints, in-flight FP8 KV-cache wire quantization, quality-guardrail validation (perplexity, cosine similarity).
- **Inference Serving:** vLLM (PagedAttention, Chunked Prefill, v1 scheduler internals), SGLang (RadixAttention, RadixCache), MoE serving (grouped GEMM, expert dispatch), speculative decoding, disaggregated prefill/decode.
- **Distributed & Parallelism:** Tensor parallelism (column/row-parallel sharding, all-reduce), asymmetric-TP KV re-sharding, NVLink/PCIe interconnect analysis.
- **Frameworks, Languages & Infra:** PyTorch, HuggingFace TRL, DeepSpeed ZeRO-3, Python, C++, Go, Kubernetes (native scheduler-plugin dev), GKE.

EDUCATION

Anna University

Bachelor of Technology (B.Tech) in Information Technology

Chennai, India

2016